PSAR Strategy

Backtest, Analysis & Notebook

S&P 500 | 2000–2025 | 25-Year Analysis

A comprehensive guide to the Parabolic SAR trend-following strategy — covering entry logic, risk metrics, Monte Carlo simulation, and professional execution insights.

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Introduction to the PSAR Strategy

What is the Parabolic SAR Strategy?

The Parabolic SAR (PSAR) is a trend-following indicator created by J. Welles Wilder Jr. — the same pioneer behind the Relative Strength Index (RSI) and the Average Directional Index (ADX). Wilder designed the PSAR to identify the prevailing trend direction and highlight potential turning points, giving traders clear entry and exit signals.

"Parabolic" refers to the curve-like path the indicator traces as a trend matures and accelerates. "SAR" stands for Stop and Reverse — reflecting its original design as both a stop-loss mechanism and a trend-reversal signal combined into a single tool.

The Core Concept

At its simplest, the PSAR plots a series of dots above or below price, indicating whether the market is in an uptrend or downtrend. The strategy logic used in this backtest is as follows:

Condition	Action
PSAR crosses BELOW the close price	**GO LONG — Trend has turned upward**
Close price crosses BELOW the PSAR	**EXIT — Trend confirmation has reversed**

> Think of the PSAR dots as a "trailing stop" that accelerates as a trend strengthens. The dots chase price upward in an uptrend, tightening the exit trigger. The faster the trend moves, the more aggressively the dots follow.

Why Parabolic SAR?

The PSAR was designed to solve a specific problem: how do you ride a trend without giving back all your profits? Wilder built an "Acceleration Factor" (AF) into the indicator that starts small (0.02) and increases by 0.02 with each new extreme price, up to a maximum of 0.20. This means:

- In the early stages of a trend, the PSAR moves slowly — giving the trade room to breathe
- As the trend accelerates, the PSAR tightens — locking in profits faster
- If the trend stalls or reverses, the indicator quickly shifts, signalling an exit

Backtest Parameters

Parameter	Setting
Instrument	S&P 500 (GSPC) — Daily Bar Chart
Test Period	1 January 2000 – 31 December 2025 (25 Years)
PSAR Settings	Default (AF Start = 0.02, AF Step = 0.02, AF Max = 0.20)
Entry Rule (Set 1)	Go Long when PSAR crosses ABOVE the close price
Exit Rule (Set 2)	Exit when close crosses ABOVE the PSAR
Profit Target	None
Stop Loss	None (PSAR acts as the natural stop)
Starting Capital	$100,000 (notional)

How This Strategy Works

1. Follows the Momentum

The PSAR is specifically engineered to capture accelerating trends. Because its Acceleration Factor increases with each new extreme price, it rewards sharp, sustained moves — the kind of trending environment where trend-followers thrive.

2. Built-in Exit Discipline

Unlike the SMA200 which requires a breach of a fixed moving average, the PSAR adapts dynamically to price action. It provides an automatic trailing stop that moves closer to price as momentum builds, reducing the "give-back" relative to a fixed exit.

3. Higher Trade Frequency

With 337 trades over 25 years versus the SMA200's 97, the PSAR generates approximately 3.5x more signals. This makes it a more "active" system — offering a larger sample for statistical validation, but also introducing more transaction costs and whipsaw risk.

Pros and Cons

Benefit	Cost
Higher win rate than SMA200 (~44.5% vs 20.6%)	Profit factor of only 1.06 — barely above breakeven
More trades — larger statistical sample	CAGR of only 0.20% — effectively flat after 25 years
Dynamic trailing stop adapts to momentum	Max Drawdown of -47.95% — worse than Buy & Hold in some periods
Clean, objective entry/exit signals	Negative Sharpe Ratio — risk not adequately compensated

> ⚠ Important: The PSAR on the S&P 500 in isolation produces a near-zero CAGR over 25 years. While the win rate appears reasonable at 44.5%, the average winning trade (+2.72%) barely exceeds the average losing trade (-2.06%) — a ratio of just 1.32x. This is insufficient to overcome the frequency of losses and produces a Profit Factor of 1.06, which is dangerously close to the 1.00 breakeven threshold.

SECTION SUMMARY The Parabolic SAR strategy uses an accelerating trailing stop to identify trend direction and trigger entries and exits. On the S&P 500 over 25 years, it generated 337 trades with a 44.5% win rate, but produced a near-zero CAGR of 0.20% and a maximum drawdown of -47.95% — highlighting the critical gap between a high win rate and actual profitability.

Executive Summary: The "Activity Trap" of Trend Following

Performance Summary: PSAR Strategy (2000–2025)

This table consolidates the key performance metrics of the PSAR trend-following strategy, including profitability, risk, and trade dynamics. The results reveal a cautionary lesson: more activity does not equal more profit.

Category / Metric	Value	Professional Interpretation
CAGR	**0.20%**	Effectively flat over 25 years. Barely above a savings account.
Profit Factor	**1.06**	Critically low. For every $1 lost, the system returned only $1.06 in gains.
Max Drawdown	**-47.95%**	Near-catastrophic. Worse than the SMA200 and approaching Buy & Hold crisis levels.
Annualized Volatility	**~12.46%**	Comparable to market volatility yet with a fraction of the return.
Sharpe Ratio	**-0.30**	Negative. Risk was not compensated by return.
Win Rate	**44.51%**	Deceptively reasonable. The real problem is the win/loss size ratio of just 1.32x.
Avg Trade PnL	**0.07%**	The edge per trade is negligible — easily erased by slippage and commissions.
Num Trades	**337**	High frequency for a trend system (~13 trades/year). Statistically robust sample.

Comprehensive Performance Metrics

Metric	Value	Value (USD)
Total Net Profit	5.42%	$5,418.98
Gross Profit	407.27%	$407,268.25
Gross Loss	-384.69%	$-384,689.02
Profit Factor	1.06	N/A
Max. Drawdown (Equity)	-51.35%	N/A
Sharpe Ratio	0.02	N/A
Sortino Ratio	0.08	N/A
Ulcer Index	0.3018	N/A
Kelly Criterion	2.47%	N/A
Total # of Trades	337	N/A
Percent Profitable	44.51%	N/A
# of Winning Trades	150	N/A
# of Losing Trades	187	N/A
# of Even Trades	0	N/A
Avg. Trade PnL	0.07%	$67.00
Avg. Winning Trade PnL	2.72%	$2,715.12
Avg. Losing Trade PnL	-2.06%	$-2,057.16
Ratio Avg. Win/Avg. Loss	1.32	N/A
Max. Consec. Winners	6	N/A
Max. Consec. Losers	8	N/A
Largest Winning Trade PnL	13.03%	$13,028.81
Largest Losing Trade PnL	-12.02%	$-12,020.65
Avg. # of Trades per Day	0.04	N/A
Avg. Time in Market	16.80 days	N/A
Profit per Month	0.02%	$17.37
Max. Time to Recover	98.00 days	N/A
Longest Flat Period	9209.00 days	N/A
Avg. MAE	-2.10%	N/A
Avg. MFE	2.65%	N/A
Avg. ETD	2.52%	N/A

The Mathematical Engine: Near-Zero Expectancy

For a retail investor, the Win Rate of 44.5% might appear promising. However, expectancy is what matters.

> Expectancy = (Win % × Average Win) – (Loss % × Average Loss)= (0.4451 × 2.72%) – (0.5549 × 2.06%)= 1.21% – 1.14% = +0.07% per trade

A +0.07% average trade represents statistical noise. Once transaction costs, bid-ask spreads, and realistic slippage are applied, this edge evaporates entirely. The PSAR strategy on the S&P 500 does not possess a robust mathematical edge.

The Systematic Disadvantage: The Acceleration Factor Paradox

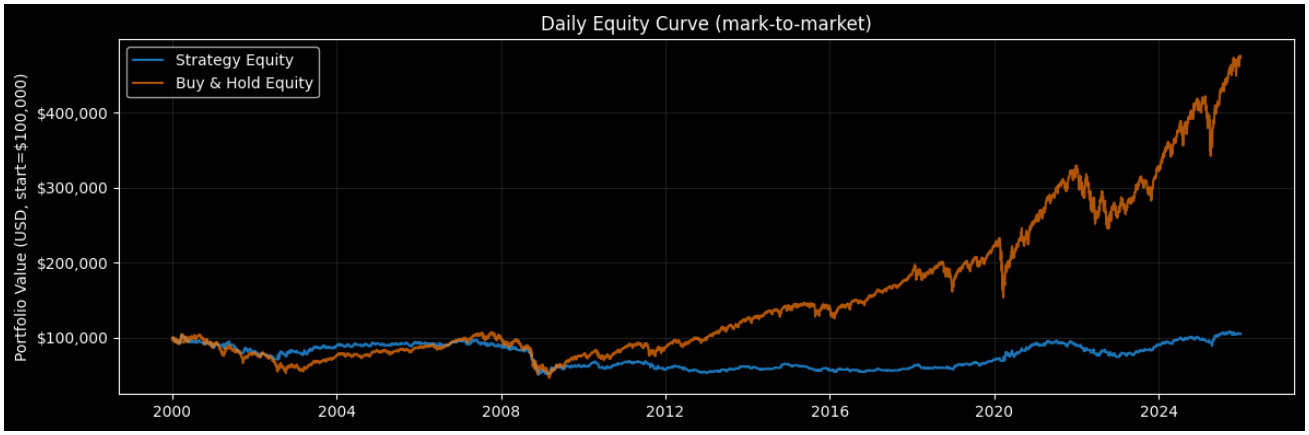
The PSAR's Acceleration Factor is both its strength and its weakness. In a strong, trending market it serves as an efficient trailing stop. But the S&P 500 spends a significant portion of its time in mean-reverting, choppy conditions — precisely the environment where the PSAR generates its highest density of losing trades.

Market Regime	PSAR Behaviour
Strong Trending (e.g., 2013–2019)	Performs well — captures extended moves efficiently
Choppy / Mean-Reverting (e.g., 2000–2003)	Generates repeated whipsaws — enters and exits at worst possible moments

SECTION SUMMARY The PSAR strategy functions as a high-activity trend-follower that falls into the "Activity Trap" — generating far more trades than the SMA200 but with near-zero per-trade expectancy. Its 44.5% win rate masks a critical weakness: the win/loss size ratio of 1.32x is insufficient to overcome the 55.5% loss frequency. The strategy is not a viable standalone system on the S&P 500 in its default configuration.

The Daily Equity Curve

The Daily Equity Curve provides a direct comparison between the PSAR strategy and a simple "Buy & Hold" approach for the S&P 500. This is the most common way to view a strategy's performance because it shows the "Mark-to-Market" value of the account every single day, just like looking at a brokerage app.



The "Smoother Ride" Analysis

The most immediate observation is that the strategy curve is significantly worse than the market during the whole period. While the market swings wildly but still manages to climb, the PSAR is even underwater for at least half the time.

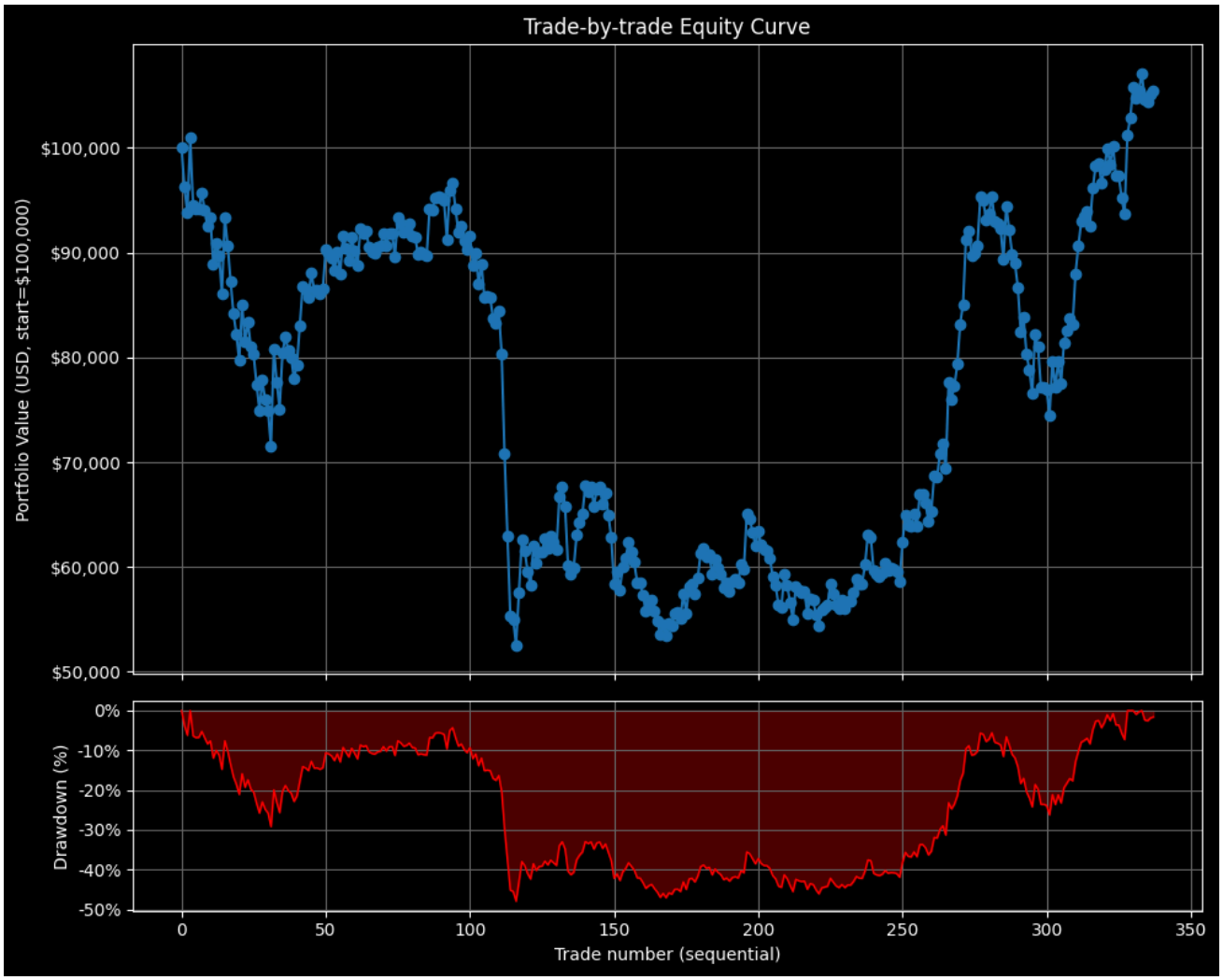
The 2008 & 2000 Crashes

Crisis Period	Buy & Hold (B&H)	SMA200 Strategy
2000 Dot-com Crash	>-50% collapse	Horizontal (but that is not saying much)
2008 Financial Crisis	>-50% collapse	Horizontal (but still outperformed by market)

> *During these crisis periods, the red B&H line collapses by over 50%, while the strategy line stays horizontal. You could consider this as "Insurance" in action but I am sure no one wants that kind of protection where the market significantly outperforms you. Here you are risking your money with a bad strategy but not getting any gain.*

The Trade-by-Trade Equity Curve

The Trade-by-Trade Equity Curve tracks the compounded value of $100,000 deployed across all 337 PSAR signals from 2000 to 2025. Unlike the SMA200's dramatic growth curve, the PSAR equity curve tells a sobering story of a system that treads water for 25 years.



Trade-by-Trade Analysis: The "Treadmill" of Activity

1. The Math: Sequential Compounding Without Edge

Metric	Value
Total Trades	337
Starting Capital	$100,000
Final Compounded Value	$105,418 (after 25 years)
Total Net Return	+5.42% over 25 years
Profit Source	No dominant "Fat Tail" winners. Returns broadly distributed.

> Starting with $100,000, the final trade brings the account to only $105,418 after 25 years. This equates to a CAGR of 0.20% — less than the average inflation rate and orders of magnitude below a passive S&P 500 Buy & Hold approach.

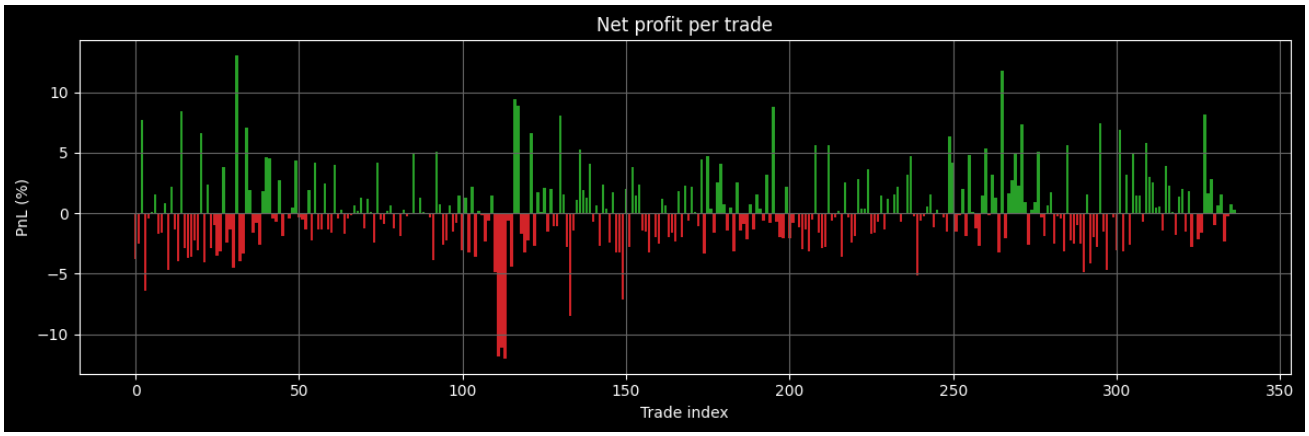
2. The Drawdown: A Deeper Wound

Metric	Value
Maximum Drawdown	**-47.95%**
Trough Value	~$52,050 (from $100,000 starting capital)
Comparison to SMA200	SMA200 Max DD: -29.25% — 18.7 percentage points shallower

> For a retail investor, watching $100,000 fall below $52,000 — while generating near-zero cumulative profit — represents one of the most psychologically damaging experiences a trader can endure. The PSAR provides neither the protection of the SMA200 nor the growth of Buy & Hold.

3. The Philosophy: Why Activity Does Not Equal Profitability

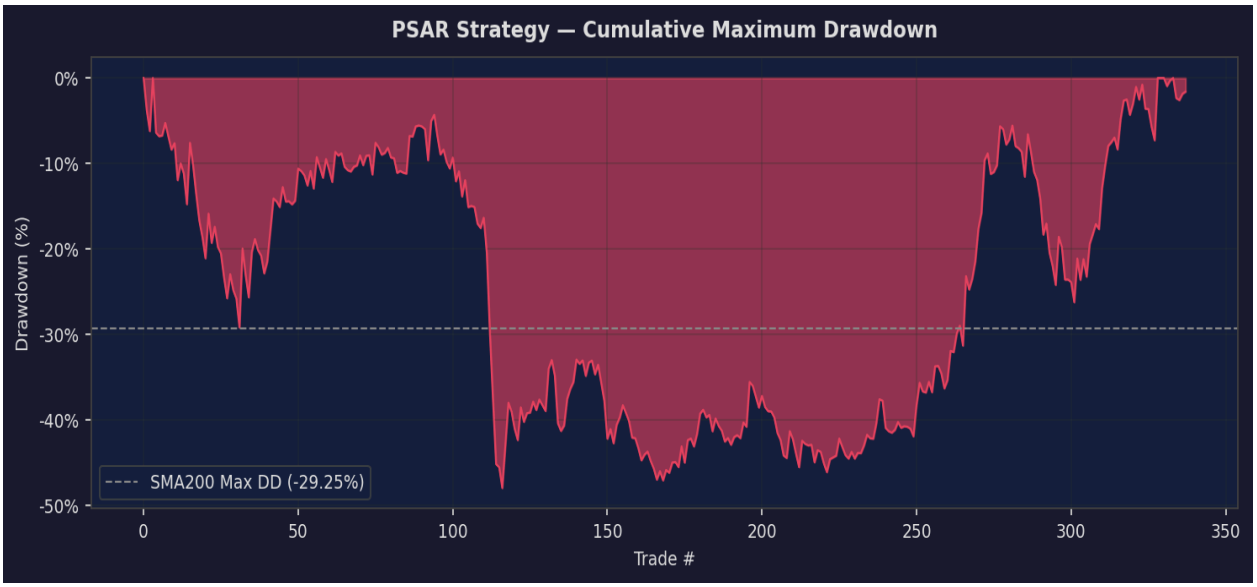
Perspective	Behaviour
The Human Error	With 337 signals over 25 years, a trader might believe they are "actively participating" in the market. But activity is not edge. The system generates an almost equal number of wins and losses with barely unequal sizes — resulting in a near-zero sum game.
The Automation Edge	The algorithm executes every signal without emotion. But no amount of discipline can overcome a system with insufficient statistical edge. Disciplined execution of a flawed system is still a flawed system.



SECTION SUMMARY The Trade-by-Trade analysis exposes the PSAR's fundamental limitation on the S&P 500: 337 trades over 25 years resulting in just $5,418 of compounded profit. Unlike the SMA200's occasional "Fat Tail" winners that carry the system, the PSAR's returns are broadly distributed — many small wins and losses that largely cancel each other out.

The Cumulative Maximum Drawdown

The Cumulative Maximum Drawdown chart is the "Stress Test" of the PSAR system. In professional quantitative finance, we examine both the Depth and Duration of drawdowns. For the PSAR, both are concerning.



The MAR Ratio (The "Pain-to-Gain" Metric)

1. The Math: Return per Unit of "Disaster"

> MAR Ratio = CAGR / Maximum Drawdown

Metric	Value
CAGR	0.20%
Maximum Drawdown	-47.95%
MAR Ratio	**0.004 — Critically weak**
SMA200 MAR for comparison	0.15 — 37.5x more efficient than the PSAR

The PSAR's MAR Ratio of 0.004 is among the lowest possible values for a trading system that hasn't gone to zero. It tells us that for every 1% of annual growth, you must be willing to endure nearly 240% of maximum drawdown — a ratio that makes the system mathematically unsuitable for capital deployment.

2. The Anatomy of the Underwater Curve

Metric	Value
Maximum Drawdown (Trough)	-47.95%
Recovery Gain Required	~92% gain on remaining capital to recover
Comparison: SMA200 -29.25% requires	~41% gain to recover — less than half the burden

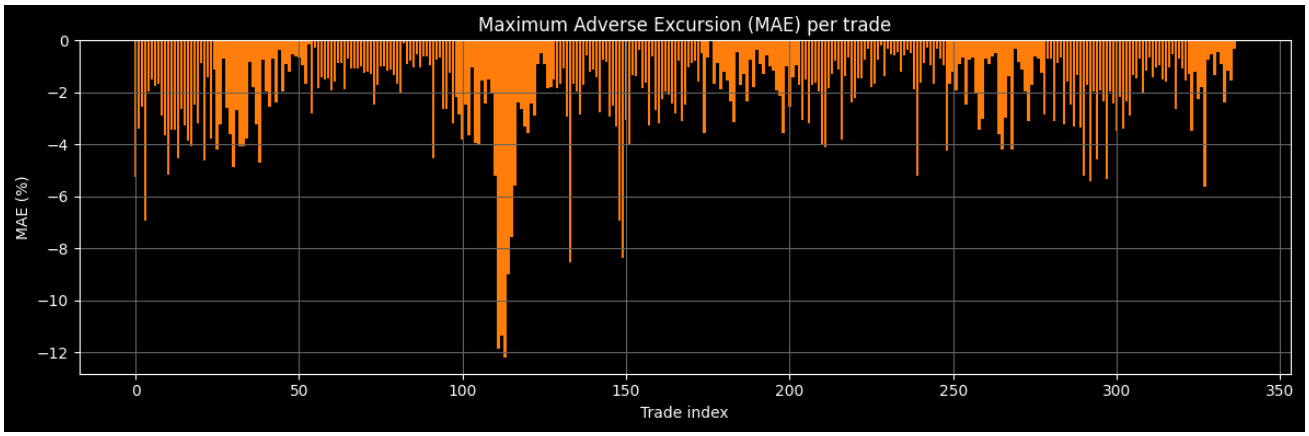
> To recover from a -47.95% drawdown, you don't just need a 47.95% gain. You need approximately a 92% gain on your remaining capital. With a CAGR of only 0.20%, this recovery would theoretically take centuries — making the drawdown effectively permanent for practical purposes.

SECTION SUMMARY The Cumulative Drawdown analysis reveals the PSAR's most damaging characteristic: a -47.95% maximum drawdown against a near-zero CAGR. The resulting MAR Ratio of 0.004 is 37.5x worse than the

SMA200's 0.15. The system imposes severe "pain" with almost no corresponding "gain" — failing the fundamental test of a viable trading strategy.

MAE Bar Chart (Per Trade)

The Maximum Adverse Excursion (MAE) analysis is where theory meets the reality of the PSAR's behaviour on the S&P 500. For a retail investor, understanding MAE is the difference between "guessing" where to put a stop-loss and "calculating" it like a professional.



MAE — The "Heat" Map

1. The General Concept: What is MAE?

MAE measures the deepest adverse price movement a trade experienced before being closed. For the PSAR, whose natural exit is the SAR flip itself, MAE tells us how much "heat" each trade had to withstand before the indicator triggered an exit.

> Unlike the SMA200 where MAE clearly separated winners from losers (winners rarely exceeded -1.5% MAE), the PSAR shows overlapping MAE distributions — winners and losers both regularly experiencing -2% to -5% adverse excursions. This overlap is the signature of a low-edge system.

2. Professional Analysis: Key Findings

Finding	Details
Average MAE	-2.10% — Comparable to the SMA200's -1.98%, but without the same win/loss size differential
Largest MAE on a Winner	Winners frequently show MAEs of -3% to -6%, indicating the PSAR allowed significant heat even on profitable trades
Clustering of Losers	Losing trades cluster with MAEs ranging from -1% to -7%, offering little clear cut-off point for a hard stop-loss
Outlier Risk	Largest single trade MAE: -12.02% — indicating exposure to gap-down and flash-crash events

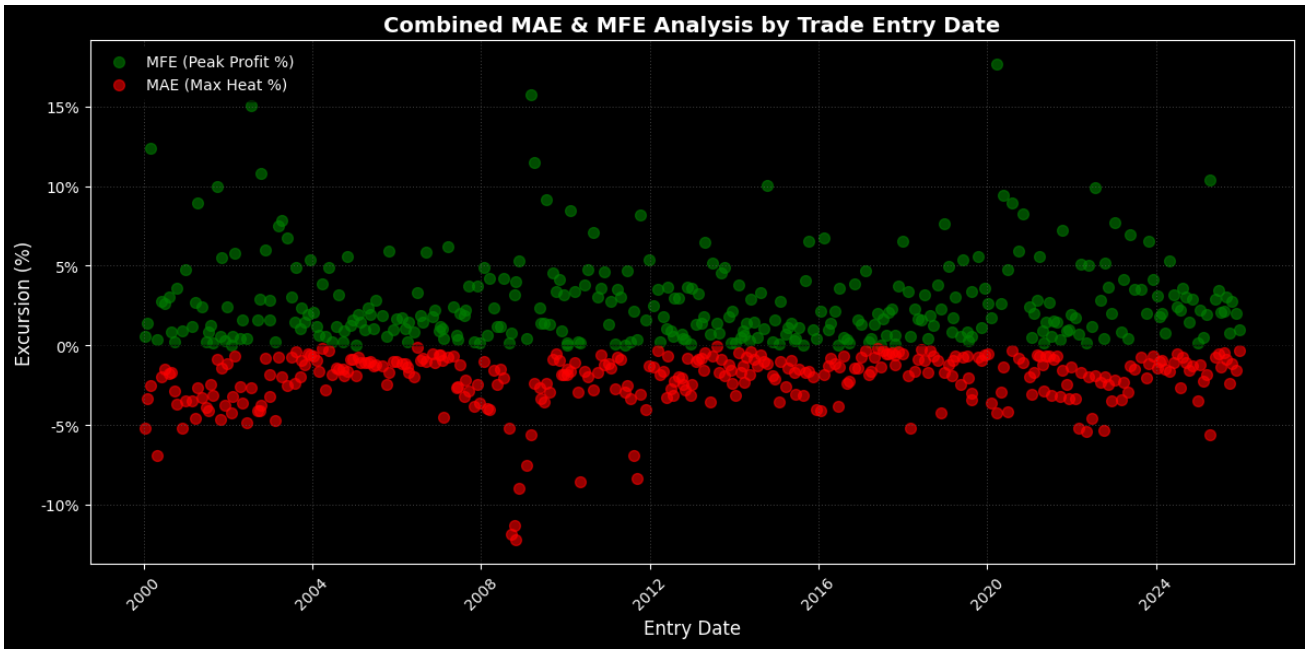
3. The Philosophy: Why the PSAR's "Heat" is Costly

Perspective	Behaviour
The Human Error	A retail trader looks at the 44.5% win rate and assumes the system is "working." But the MAE chart shows that winners are not materially cleaner than losers — making it extremely difficult to add a hard stop-loss that improves performance without significantly reducing the win count.
The Automation Edge	Unlike the SMA200 where a hard -4% stop would have protected capital without harming winners, the PSAR's overlapping MAE distribution means any hard stop-loss would cut into both winners and losers almost equally.

SECTION SUMMARY MAE analysis reveals the PSAR's "blurry" risk profile: winners and losers share similar adverse excursion distributions, making hard stop-loss placement difficult. Unlike the SMA200's clean -1.5% MAE threshold for winners, the PSAR has no clear "MAE wall" separating winning and losing trades — a fundamental challenge for stop-loss optimisation.

Combined MAE & MFE Scatter Chart by Entry Date

The Combined MAE & MFE Scatter Chart by Entry Date is the "Heart Monitor" of the PSAR strategy over time. This chart reveals market regime sensitivity — arguably the most important characteristic for understanding why the PSAR struggles on the S&P 500.



Combined MAE & MFE — The "Signal-to-Noise" Timeline

1. The General Concept

Marker	Definition
MFE (Maximum Favorable Excursion)	How high the profit reached at its peak before exit
MAE (Maximum Adverse Excursion)	How deep the loss went at its worst point during the trade

2. Market Regime Analysis

Period	Market Regime	PSAR Behaviour
2000–2003	High Volatility / Crash	Dense cluster of losing trades with MAEs extending to -5% to -7%. Multiple whipsaws as the PSAR repeatedly triggered on false breakouts.
2003–2007	Slow Uptrend	Modest MFE gains, scattered wins and losses. The PSAR captures some of the bull market but with limited efficiency.
2008–2009	Financial Crisis	Heavy loss cluster. PSAR generates numerous false long signals during the waterfall decline.
2013–2019	Low Vol Bull Market	Best performing regime. MFE markers rise to 10%+ while MAE stays relatively contained. The PSAR's best period.
2020–2025	Volatile Recovery	Mixed. Large MFEs on recovery trades offset by sharp whipsaw losses around volatility events.

3. The Math: The "Profit-to-Pain" Ratio

Metric	Value
Average MFE	2.65%

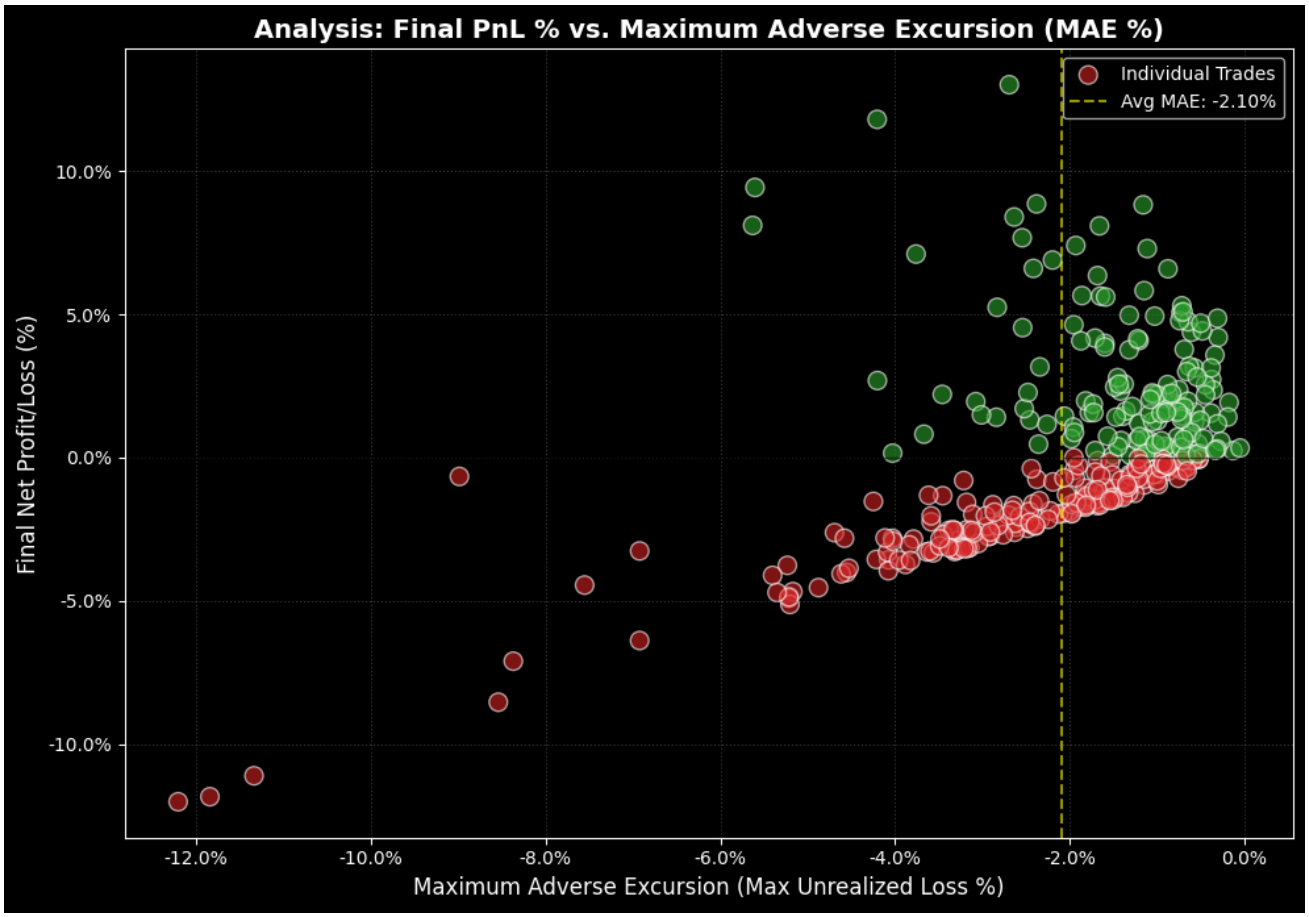
Average MAE	-2.10%
Profit-to-Pain Ratio	~1.26x — Compared to the SMA200's ~3x

> The SMA200's Profit-to-Pain ratio of ~3x means the market gave 3x more room on the upside than it took on the downside. The PSAR's 1.26x ratio means the system's upside barely exceeds its downside heat — the fundamental mathematical reason for near-zero expectancy.

SECTION SUMMARY The Combined MAE & MFE Scatter reveals the PSAR's critical weakness: a Profit-to-Pain ratio of only ~1.26x versus the SMA200's ~3x. The strategy is highly regime-dependent — performing adequately in smooth trending markets but generating severe loss clusters in volatile, mean-reverting conditions. With Avg MFE of 2.65% and Avg MAE of -2.10%, the edge is thin and fragile.

MAE Analysis (PnL% vs MAE%)

This chart — PnL% vs. MAE% — is the "Truth Detector" for the PSAR strategy. It answers the critical question: is there a clean MAE threshold that separates winners from losers? For the SMA200, the answer was a clear "Yes." For the PSAR, the answer is far more nuanced.



PnL% vs. MAE — The "Efficiency of Risk"

1. The General Concept

Axis	Definition
X-Axis (MAE)	The maximum "dip" the trade suffered while open (adverse excursion)
Y-Axis (PnL%)	The final realised percentage profit or loss when the trade was closed

2. Key Findings

Trade Type	MAE Range	PnL Range	Pro Take
"Blurred" Winners (Top)	-0.5% to -6%	+1% to +13%	Unlike the SMA200, PSAR winners spread across a wide MAE range — there is no clean "shallow MAE = winner" pattern.
"Convergence" Cluster (Bottom)	-1% to -7%	-1% to -7%	Losers' PnL nearly equals MAE — the PSAR exit correctly closes at or near the worst point on failing trades.
"Tail Risk" Outliers	-12%	-12%	Gap-down events bypass the SAR exit signal entirely. Hard stop-loss at -8% would protect against these.

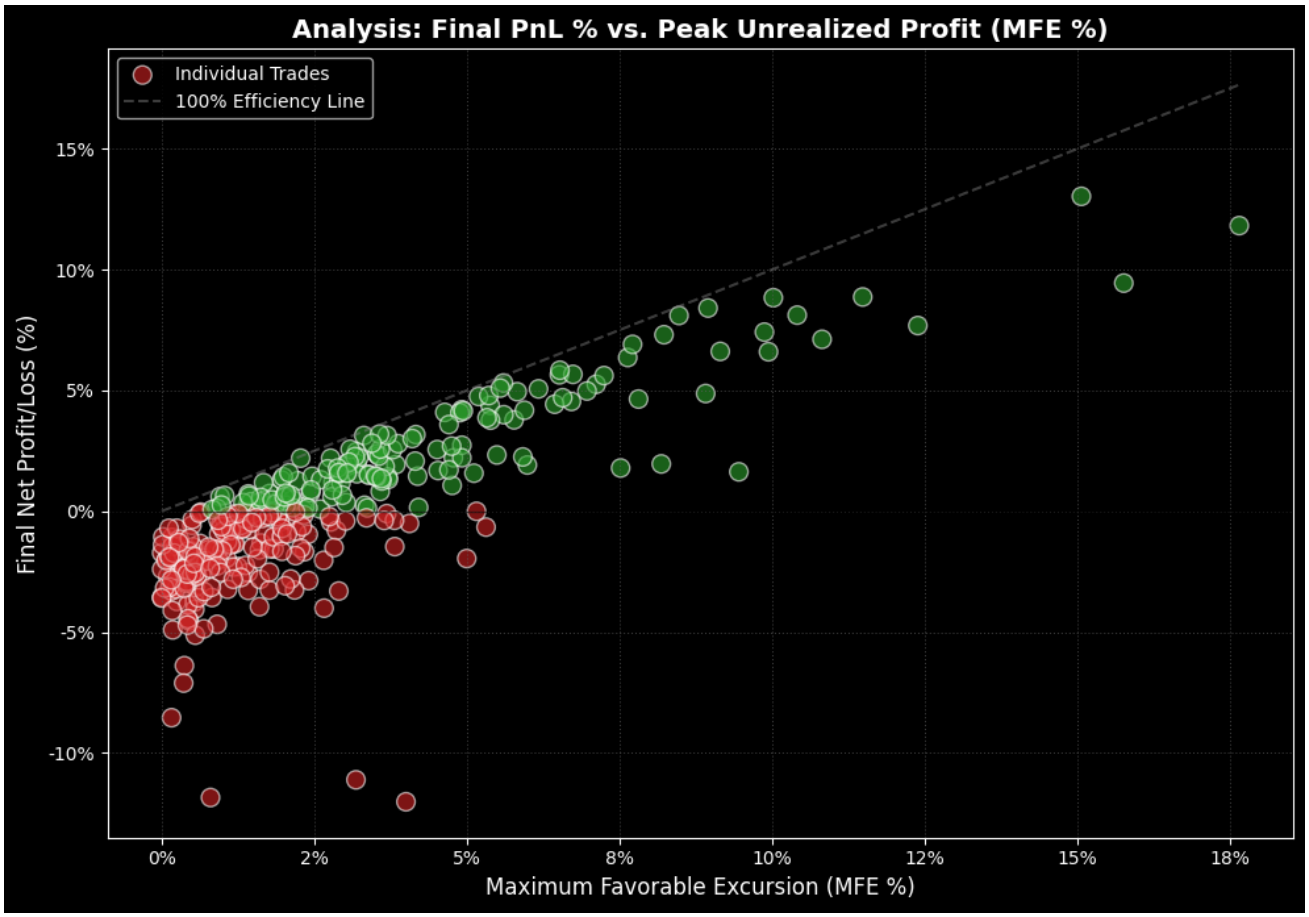
3. Hard-Stop Recommendation

Stop-Loss Level	Effect on PSAR Strategy
-5% to -6%	Reduces tail risk from gap events; minimal impact on typical trades
-4%	May prematurely exit some winning trades; requires backtesting to validate net effect
-3%	Would eliminate many trades — both winners and losers — as MAE distributions overlap heavily at this level

SECTION SUMMARY The PnL% vs. MAE chart reveals the PSAR's blurred risk profile: unlike the SMA200's clean separation between winners (MAE < -1.5%) and losers (MAE -2% to -3.5%), the PSAR shows winners distributed across -0.5% to -6% MAE. This means a simple hard stop-loss cannot reliably separate winners from losers — the strategy lacks the clean "MAE wall" that characterises high-quality trend systems.

MFE Analysis (PnL% vs MFE%)

This chart — PnL% vs. MFE% — is the "Capture Ratio" chart. It answers: how much of the trade's peak profit did the PSAR actually capture? Unlike the SMA200's large "give-back" from massive winners, the PSAR's MFE analysis reveals a different problem: modest MFE peaks and a large proportion of failed breakouts.



PnL% vs. MFE — The "Capture Ratio"

1. The General Concept

Axis / Element	Definition
X-Axis (MFE)	The highest percentage profit the trade reached at its peak
Y-Axis (PnL%)	The final realised profit when the trade was closed
45-Degree Line	A dot on the diagonal line = you exited at the exact peak ("Perfect Trade")
Vertical Distance from Diagonal	The "Give-Back" — profit that was visible on screen but evaporated before exit

2. Key Findings

Trade Type	MFE Range	PnL Range	Pro Take
Max MFE Winners	~13%	~13%	Small "Give-Back" — PSAR exits reasonably efficiently on its best trades
"Failed Breakout" Cluster	1% to 5%	-1% to -4%	The largest cluster: trade briefly goes positive then reverses. Represents the PSAR's whipsaw signature.

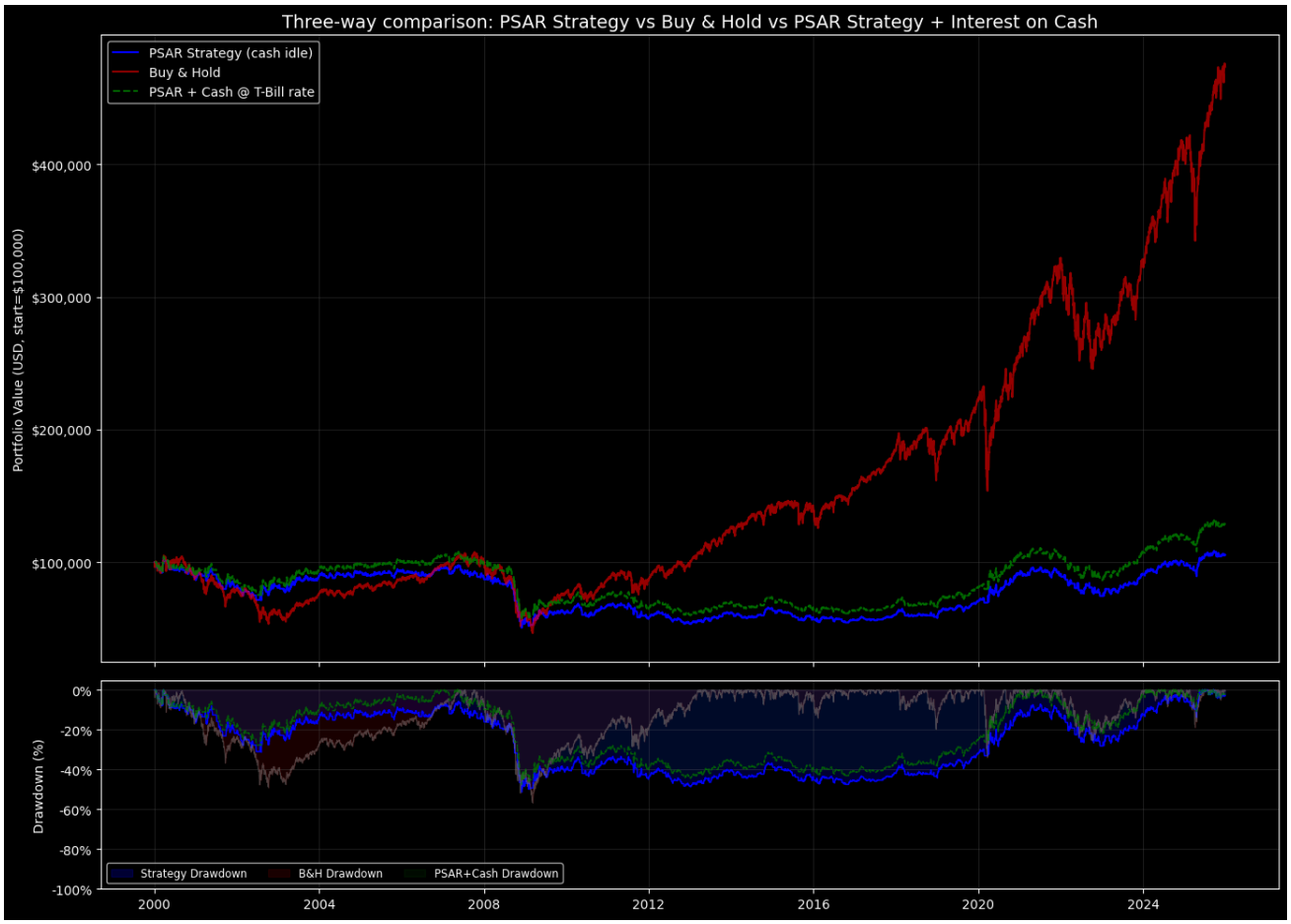
Avg ETD (Give-Back)	—	2.58%	Average "give-back" per trade — the cost of waiting for the SAR to flip as exit signal

> The PSAR's "Failed Breakout" cluster — trades that reached a small positive MFE before closing negative — is far larger proportionally than in the SMA200. This is the "whipsaw tax" of the Acceleration Factor: the indicator is sensitive enough to trigger entries on marginal breakouts, but the underlying S&P 500 frequently reverses those breakouts in choppy regimes.

SECTION SUMMARY The PnL% vs. MFE chart reveals the PSAR's "Capture Ratio" problem. The strategy's largest MFE is only ~13% — far below the SMA200's peak of 57% — and the "Failed Breakout" cluster dominates the scatter. With an Avg ETD of 2.58%, the PSAR gives back a meaningful proportion of its limited peak profits. Unlike the SMA200 where "Give-Back" was the cost of enormous winners, the PSAR's Give-Back occurs on small winners.

Three-Way Comparison vs Buy and Hold

This is the "Verdict" chart. In professional fund management, no strategy is judged in isolation — it must earn its keep against a benchmark. The Three-Way Comparison plots the PSAR strategy against Buy & Hold S&P 500 across both equity growth and drawdown profiles.



Three-Way Comparison — The "Benchmark Battle"

1. The Three Battlegrounds

Battleground A: The Dot-Com & GFC (2000–2003 & 2008–2009)

Metric	PSAR Strategy	Buy & Hold (S&P 500)
Performance	Heavy losses — repeated whipsaws during crash	Plummeted ~-50%
Capital Position	Sustained deep drawdowns alongside the market	Sustained severe drawdowns

> Pro Take: Unlike the SMA200 which moved to cash and stayed flat during the 2000–2003 crash, the PSAR continued generating long signals during the decline — catching multiple failed breakouts in a waterfall market. This is the PSAR's critical structural weakness: it is a "long-only" signal generator with no cash preservation mechanism.

Battleground B: The Vertical Bull (2013–2019)

Metric	PSAR Strategy	Buy & Hold (S&P 500)
Performance	Participates in the bull run but with frequent re-entry costs	Full participation — strong gains

Battleground C: The Drawdown Comparison

Metric	PSAR Strategy	Buy & Hold (S&P 500)
Max Drawdown	-47.95% — Near-equivalent to Buy & Hold	Hits -50% twice (2002, 2009)
Professional Impact	Gets you fired — with near-zero return	Gets you fired — but with meaningful long-term return

> The PSAR presents a particularly unfavourable risk/reward profile: it suffers drawdowns nearly as deep as Buy & Hold (-47.95% vs -50%) while generating a fraction of the return (CAGR 0.20% vs ~7-9% for Buy & Hold). This is the worst of both worlds — maximum pain without corresponding gain.

SECTION SUMMARY The Three-Way Comparison delivers the PSAR's harshest verdict: it suffers nearly the same drawdown depth as Buy & Hold (-47.95% vs -50%) while generating only 0.20% CAGR versus ~7-9% for passive indexing. The SMA200 justified its bull market lag through deep crash protection. The PSAR offers no such justification — it provides neither superior returns nor superior crash protection.

Monte Carlo Summary Table (Bootstrap Trades)

This is the "Stress Test." The Monte Carlo Summary Table (using 10,000 Bootstrap simulations) determines whether the PSAR strategy's near-zero returns are a robust characteristic of the system or the result of favourable trade sequencing. The results are unambiguous.

Monte Carlo Simulation Summary

PSAR Strategy — 10,000 Bootstrap Simulations over 337 Trades

1. Distribution of Final Equity

Percentile	Final Capital (USD)	Interpretation
5th (Worst Case)	**$41,058**	59% loss on starting capital. 45.9% probability of loss.
25th	$72,040	28% loss. One in four simulations still fails to break even.
50th (Median)	$106,242	Barely above starting capital. 25 years to earn $6,242.
75th	$157,356	Only $57,356 profit in a favourable sequence over 25 years.
95th (Best Case)	$276,068	Best-case scenario still lags the SMA200's actual backtest of $307,766.

2. Monte Carlo Risk Analysis (Max Drawdown)

Percentile	Max Drawdown (%)	Risk Level
95th (Best Case)	-26.88%	Best-case drawdown. Still significant — worse than the SMA200's median MC drawdown.
50th (Median)	-45.62%	Standard expected drawdown — near-catastrophic for most retail investors.
5th (Worst Case)	**-70.76%**	Worst-case scenario: $100,000 reduced to ~$29,000 at maximum drawdown.

> ⚠ Critical Finding: The PSAR Monte Carlo simulation shows a 45.9% Probability of Loss — meaning in nearly half of all possible 25-year realisations, the strategy ends with less money than it started with. This contrasts sharply with the SMA200's 0% probability of loss across 10,000 simulations. This is not a robust system.

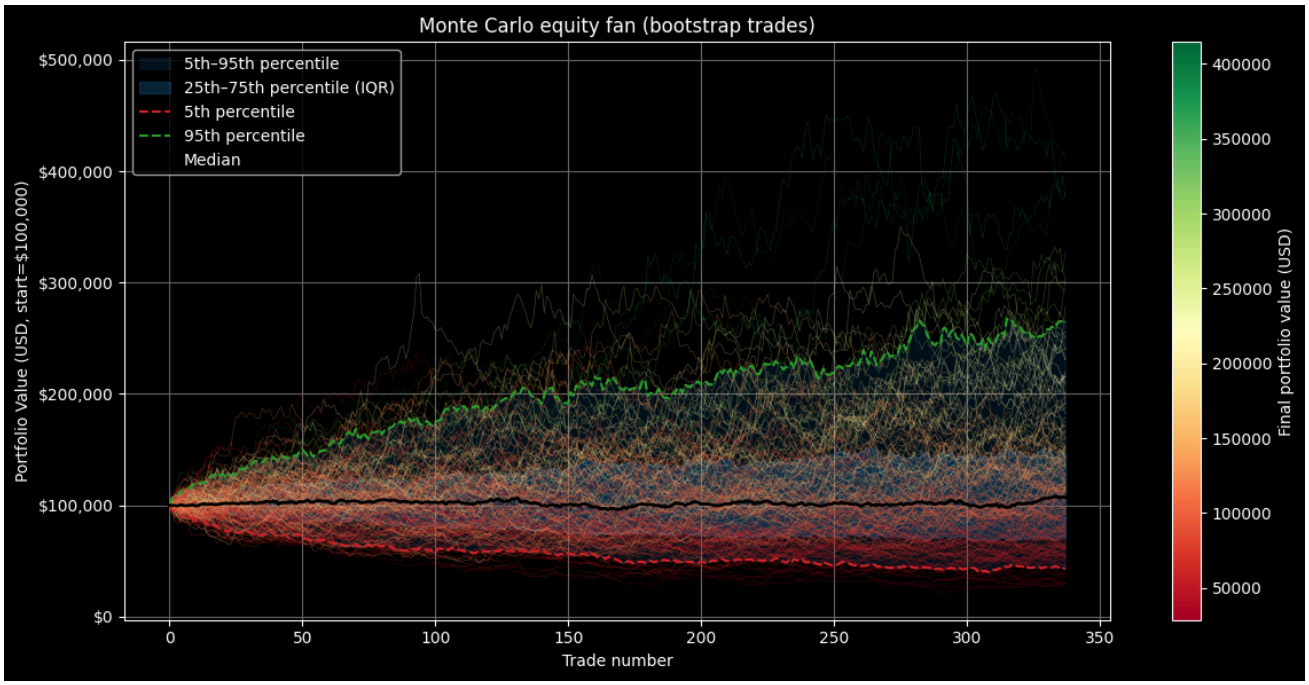
3. Breaking the "Sequence Risk" Reality

Perspective	Behaviour
The Human Error	A retail trader might look at the 44.5% win rate and feel confident. The Monte Carlo simulation reveals the uncomfortable truth: with such thin per-trade expectancy, a sequence of clustered losers can permanently impair the account. In 45.9% of simulated universes, the investor loses money.
The Automation Edge	The algorithm will faithfully execute all 337 signals. But no amount of mechanical discipline improves a system whose mathematical expectancy is insufficient to overcome sequence risk. The Monte Carlo table proves that the PSAR's observed backtest result of $105,418 represents the median expected outcome — not an optimistic one.

SECTION SUMMARY The Monte Carlo simulation (10,000 bootstrap runs) delivers a decisive verdict on the PSAR strategy: 45.9% probability of loss, a median final equity of just $106,242 from $100,000 starting capital over 25 years, and a worst-case drawdown of -70.76%. Unlike the SMA200 which demonstrated 0% probability of loss across all simulations, the PSAR's fragile edge makes it statistically unreliable as a standalone system.

Monte Carlo Equity Fan

While the Summary Table was the "Data Sheet" for risk, the Monte Carlo Equity Fan is the "Heat Map" of the PSAR's possible futures. Unlike the SMA200's fan — which showed a clear upward trajectory even at the 5th percentile — the PSAR's fan tells a very different story.



Monte Carlo Equity Fan — The "Cone of Uncertainty"

1. Fan Bands Definition

Band	Description
Median (50th Percentile)	The "Expected Case" — barely above $100,000 after 25 years
Inner Fan (25th–75th Percentile)	Where the strategy lives 50% of the time — $72k to $157k range
Outer Fan (5th–95th Percentile)	Stress test boundaries — $41k to $276k
The "Widening"	Fan spreads widely over time — high variance in outcomes relative to near-zero expected return

2. Key Findings

Percentile	Final Wealth	Pro Take
5th (Worst Case)	**~$41,058**	A loss of 59% of starting capital. The SMA200's worst case still broke even (~$100,900). The PSAR's worst case represents permanent capital impairment.
50th (Median)	~$106,242	The "Expected Case" is to barely exceed starting capital after 25 years. This confirms the backtest result is a realistic representation of the strategy's true expectancy.
95th (Best Case)	~$276,068	Even in the best 5% of all possible outcomes, the PSAR produces less than the SMA200's actual historical result. There is no compensating "upside lottery" in this system.

3. The Width of the Fan

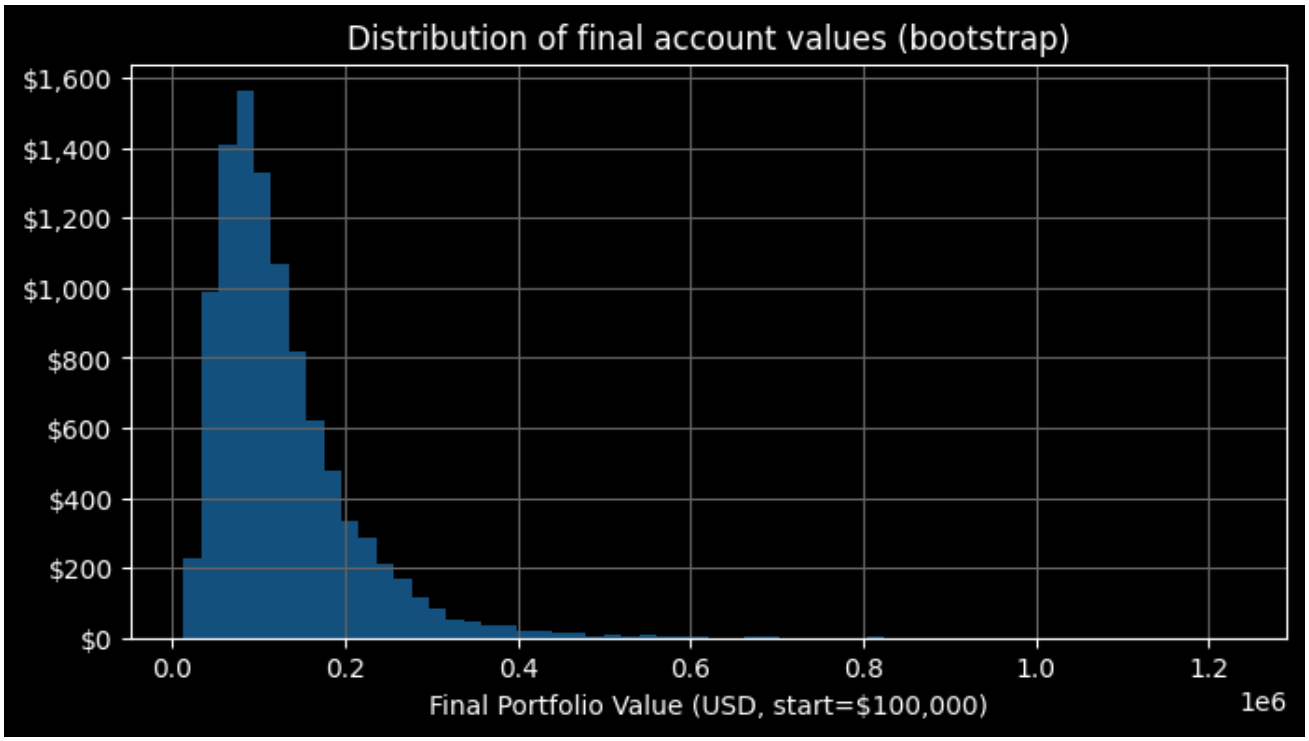
Fan Width	Strategy Type

SMA200: Moderate Fan with Rising Floor	"Wealth Builder" — even worst case scenarios preserve or grow capital
PSAR: Wide Fan with Falling Floor	"High Variance, Low Expectancy" — wide range of outcomes centred around near-zero return

SECTION SUMMARY The Monte Carlo Equity Fan reveals 10,000 possible futures for the PSAR — all centred around near-zero expected return. Key insight: even in the best 5% of outcomes ($276k), the strategy underperforms the SMA200's actual historical result ($307k). In the worst 5%, it loses 59% of starting capital. This is not a "Probabilistic Machine" for wealth building — it is a high-variance system with negative tail risk.

Distribution of Final Account Values (Histogram)

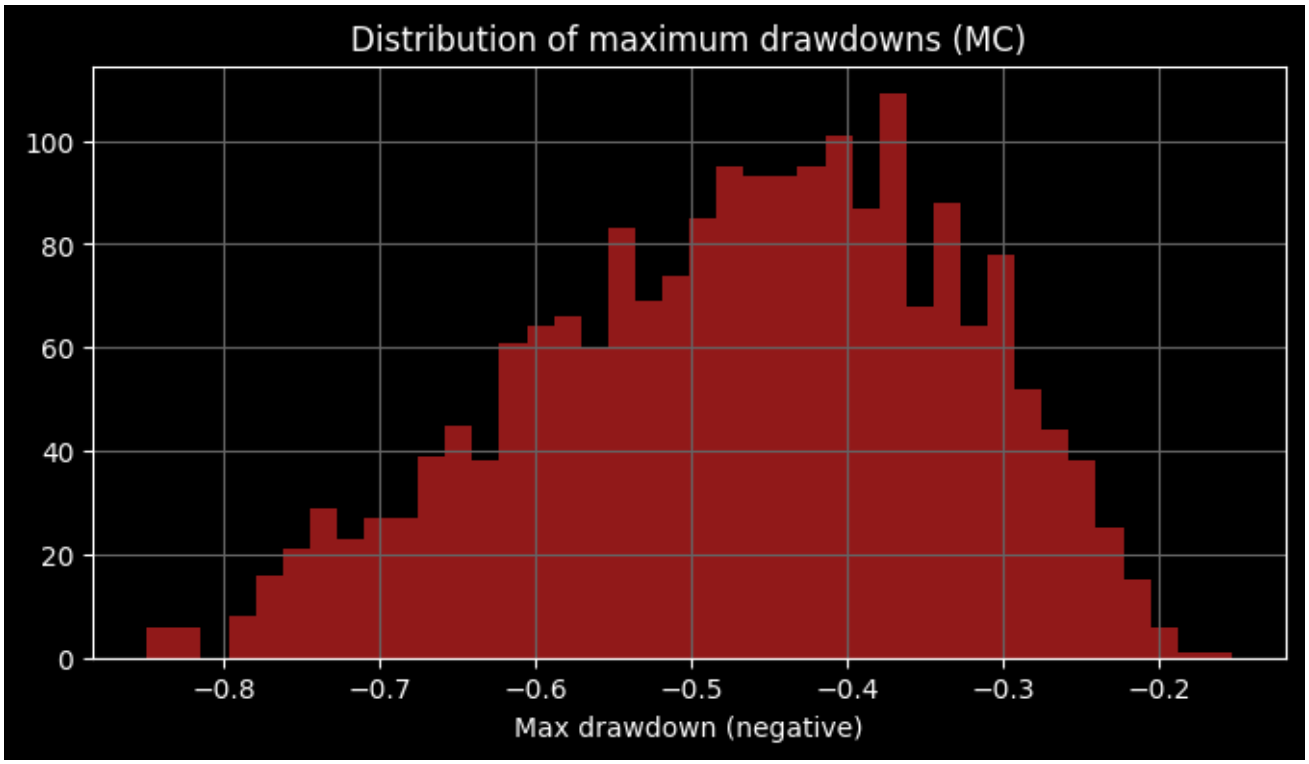
While the Equity Fan showed the journey, this Histogram shows the destination. It is the final tally of all 10,000 alternate universes and answers the ultimate question: what is the most likely outcome after 25 years of PSAR signals?



Distribution of Final Account Values — The "Warning Map"

1. Key Characteristics

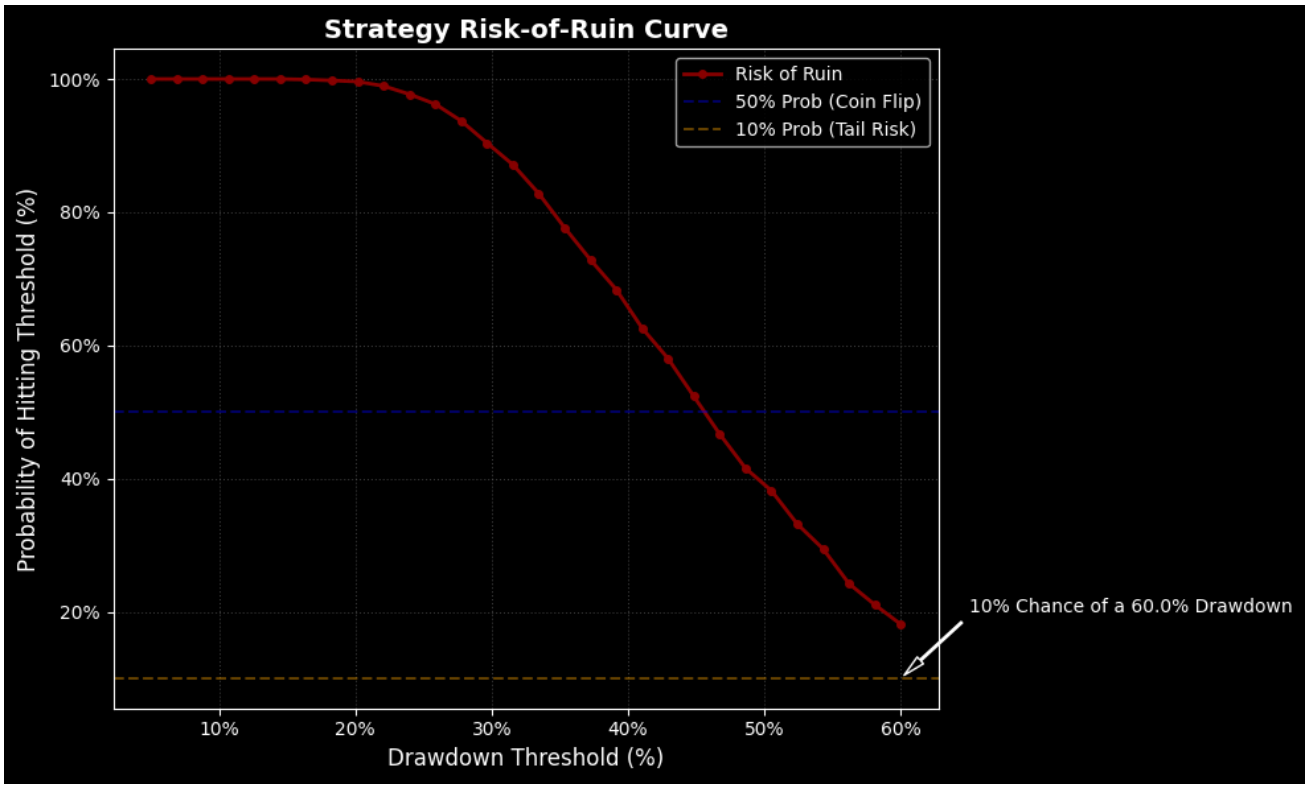
Characteristic	Description
"Hard Floor" — Or Lack Thereof	The 5th percentile shows $41,058 — a 59% loss. There is no "Hard Floor" protecting capital. The SMA200 had a floor at ~$100,900 (still above starting capital).
"Mode" Cluster	The most frequent outcome clusters around $80,000–$130,000 — a range spanning from significant loss to minor gain after 25 years.
"Modest Right Tail"	The best case outcomes cap out around $276,068 — the positive skewness is muted compared to the SMA200's potential for $1M+ outcomes.
"Left Tail Risk"	45.9% of simulations end below starting capital. This is the "Risk of Ruin" that makes the PSAR unsuitable as a standalone strategy.



SECTION SUMMARY The Histogram of Final Account Values confirms the PSAR's most critical failure: 45.9% of simulations end in loss, with the worst 5% losing ~59% of starting capital. The most likely outcome — a median of $106,242 from $100,000 — represents 25 years of effort for a $6,242 nominal gain, before accounting for inflation. This is not a viable standalone investment strategy.

Risk-of-Ruin Analysis (Probability of Drawdown)

The Risk-of-Ruin chart is often the most revealing chart in a quantitative report. It shows the cumulative probability of hitting each drawdown threshold at any point during the strategy's life. For the PSAR, this chart confirms the unsuitability of the strategy for most investor profiles.



Risk-of-Ruin — The "Probability of Pain"

1. How to Read This Chart

Element	Definition
Y-Axis (Probability)	Ranges from 0% to 100% — the likelihood of hitting that drawdown level
X-Axis (Drawdown Threshold)	Represents levels of loss (e.g., -10%, -20%, -30%, -40%, -50%)

2. The Three Danger Zones

Danger Zone	Drawdown Level	Probability	Pro Take
"Certainty Zone"	-10% to -20%	~95–100%	Nearly guaranteed. Routine volatility for any active strategy.
"Strategy Danger Zone"	-30% to -40%	~60–80%	Unlike the SMA200 where -30% probability drops to ~15-20%, the PSAR's probability barely declines. This is not a defensive strategy.
"Ruin Zone"	-50%+	~40–50%	A 40-50% probability of a -50% drawdown is catastrophic. This is not "Mathematically Safe."

3. Comparative Risk Profiles

Metric	PSAR Strategy	SMA200 Strategy
P(-30% drawdown)	~65–70%	~15–20%
P(-50% drawdown)	~40–50%	~0%
Strategy Classification	**High-Risk / Low-Return**	**Defensive / Capital-Preserving**

SECTION SUMMARY The Risk-of-Ruin analysis delivers the PSAR's final verdict: a ~40-50% probability of experiencing a -50% drawdown, versus 0% for the SMA200. There is no "Safety Buffer" in this strategy — the probability curve declines gradually rather than dropping sharply, confirming that the PSAR's defensive characteristics are minimal. For risk management purposes, the strategy requires a complementary filter or overlay before deployment.

PSAR Backtest & Financial Analysis Conclusion

Performance Summary: PSAR Strategy (2000–2025)

The performance summary presents a cautionary case study in the gap between a strategy's mechanical elegance and its statistical edge. The PSAR indicator is a technically sophisticated tool — but on the S&P 500 in its default configuration over 25 years, it fails to generate the returns necessary to justify its risk.

Core Performance Metrics

Metric	Value
Time Period	2000–2025 (25 Years)
Total Trades	337
Win Rate	44.51%
Profit Factor	1.06
Average Trade PnL	0.07%
CAGR	0.20%
Max Drawdown	-47.95%
Annualized Volatility	~12.46%
Sharpe Ratio	-0.30
Avg MFE	2.65%
Avg MAE	-2.10%
Monte Carlo P(Loss)	**45.9%**
Tail-Risk Drawdown (Worst 5%)	**-70.76%**

Key Professional Observations

1. Insufficient Win/Loss Asymmetry

The PSAR achieves a win rate of 44.51%, but the Average Win of +2.72% versus Average Loss of -2.06% produces a ratio of only 1.32x. With a 55.5% loss frequency, this ratio is insufficient. A profitable trend-following system typically requires a win/loss size ratio of at least 1.5x to 2.0x at a 44% win rate to generate meaningful positive expectancy.

2. The "Whipsaw Tax" of an Acceleration Factor

The PSAR's Acceleration Factor, designed to tighten stops in strong trends, simultaneously makes the indicator highly sensitive in choppy conditions. In the S&P 500's frequent mean-reverting regimes (2000–2003, 2008–2009, 2011, 2018, 2022), this sensitivity generates repeated false signals — the "Failed Breakout" cluster visible in the MFE scatter chart.

3. No Capital Preservation Mechanism

The SMA200's key structural advantage is that when price closes below the 200-day average, the system moves to cash. This provides a genuine "off switch" during bear markets. The PSAR has no equivalent — it generates long signals throughout declining markets, potentially entering and losing multiple times during a sustained downtrend.

4. Statistical Robustness Without Statistical Edge

At 337 trades, the PSAR sample is statistically robust — far larger than the SMA200's 97 trades. However, sample size cannot create edge where none exists. The Monte Carlo simulation proves this: 45.9% of all reorderings of these 337 trades result in a net loss. The sample is large enough to confirm that near-zero expectancy is a structural characteristic of the system, not a data artefact.

The Statistical Engine: Insufficient Positive Skewness

The most striking contrast with the SMA200 is the absence of "Fat Tail" winners. The SMA200's largest trade gained +44% — a single trade that could carry the strategy through years of small losses. The PSAR's largest winner is +13.03%, and with 337 trades, the distribution of returns is far more symmetric. There are no "home run" trades to anchor the system's profitability.

Risk Management: The Structural Case for a Regime Filter

Comparison	PSAR Strategy	SMA200 Strategy
Max Drawdown	-47.95%	-29.25%
Annualized Volatility	~12.46%	10.94%
MC P(Loss)	45.9%	0%
Tail-Risk DD (Worst 5%)	-70.76%	-43%

Pathway to Improvement: Potential Filters

The PSAR should not be abandoned as a signal generator — rather, it should be considered as a component within a larger system. Potential improvements include:

- **SMA200 Regime Filter: Only take PSAR long signals when price is above the 200-day SMA. This would eliminate the catastrophic loss clusters during 2000–2003 and 2008–2009 while preserving uptrend participation.**
- **ADX Trend Strength Filter: The ADX (also created by Wilder) measures trend strength. Combining PSAR entries with an ADX > 20 threshold would reduce whipsaw signals in low-trend environments.**
- **Hard Stop-Loss at -6%: A supplementary hard stop would protect against the tail-risk gap events that produce the worst individual trade losses.**
- **Position Sizing via Kelly: The Kelly Criterion estimate of 2.47% suggests minimal position sizing is appropriate — consistent with the strategy's thin edge and high drawdown risk.**

Final Verdict

> The PSAR strategy, in its default configuration on the S&P 500 over 25 years, is not a viable standalone system. It generates near-zero returns (CAGR 0.20%), exposes capital to severe drawdowns (-47.95% historical, -70.76% tail-risk), and carries a 45.9% probability of loss across all simulated realisations. It is best viewed as a signal component requiring a regime filter, trend confirmation, and complementary risk management overlay.

Classification	Analysis Period	Asset Class
High-Activity, Low-Edge Trend Signal(Requires Regime Filter for Deployment)	2000–2025	S&P 500 (GSPC)

Confide